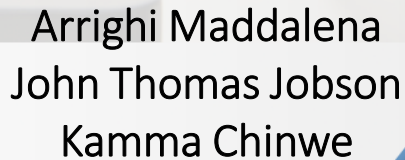


A composite image at the top of the slide. On the left, a blurred image of a microscope. On the right, a close-up of a hand in a white glove holding a test tube containing a red liquid. A blue diagonal line separates the two images.

CHE-494: Biopharmaceutical Process Development and Manufacturing

The names of the project team members are listed in a vertical stack.

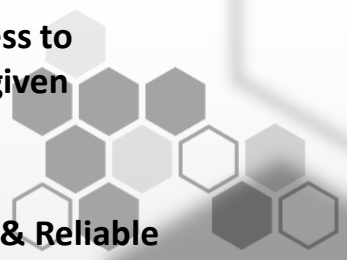
Project Team: 1

Arrighi Maddalena
John Thomas Jobson
Kamma Chinwe





CONTENTS

- 01 | Who we are
 - 02 | Drug Product Process for a Low Bulk Density and Poor Flowability Drug Substance
 - 03 | Reliable scale-up process
 - 04 | Importance of Drug Substance Properties in Drug Product Process for Solid Dosage Form
 - 05 | **Crystallization:** most prevalent unit operation in small molecule drug substance process
 - 06 | **Lyophilization:** the most critical unit operation in large molecule drug product
 - 07 | **Crystallization process to maximize the yield given solubility values**
 - 08 | **Cell Culture Process & Reliable Scale Up**
- 

01 Passion for Biopharmaceutical Companies

Chinwe Kamma



- Spark: LCA 
- Drive: Passion
- Experience: Biologics transport through mucus/Amgen
- Goal: 

Maddalena Arrighi



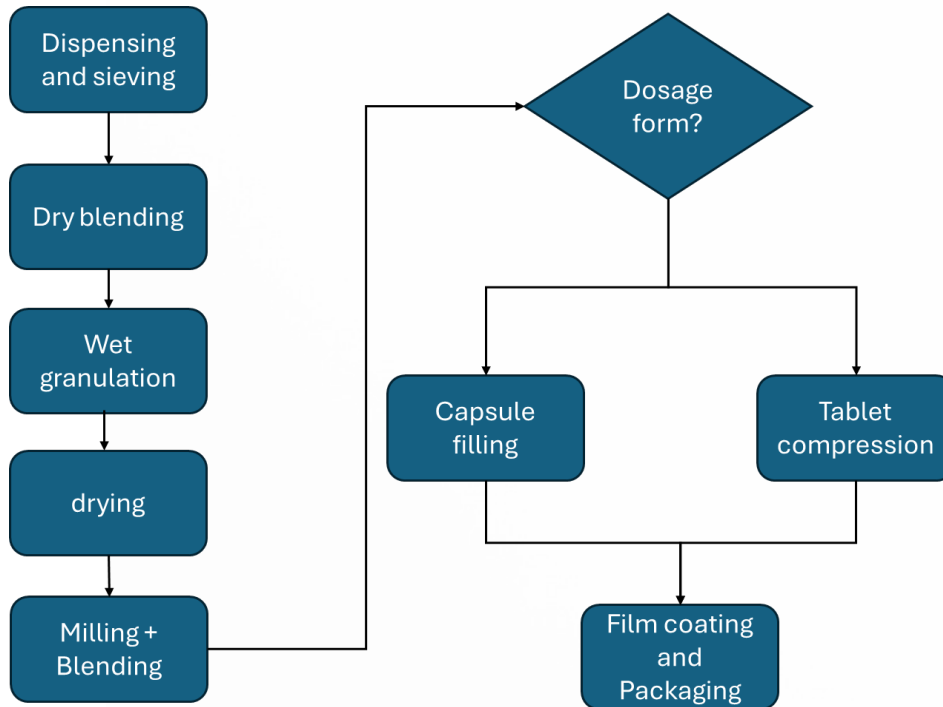
- Spark: 
- Drive: 
- Experience: microfluidics & 3D spheroids
- Goal: 

Jobson



- Spark: 
- Drive: 
- Experience: 
- Goal: 

Drug Product Process for a Low Bulk Density and Poor Flowability Drug Substance



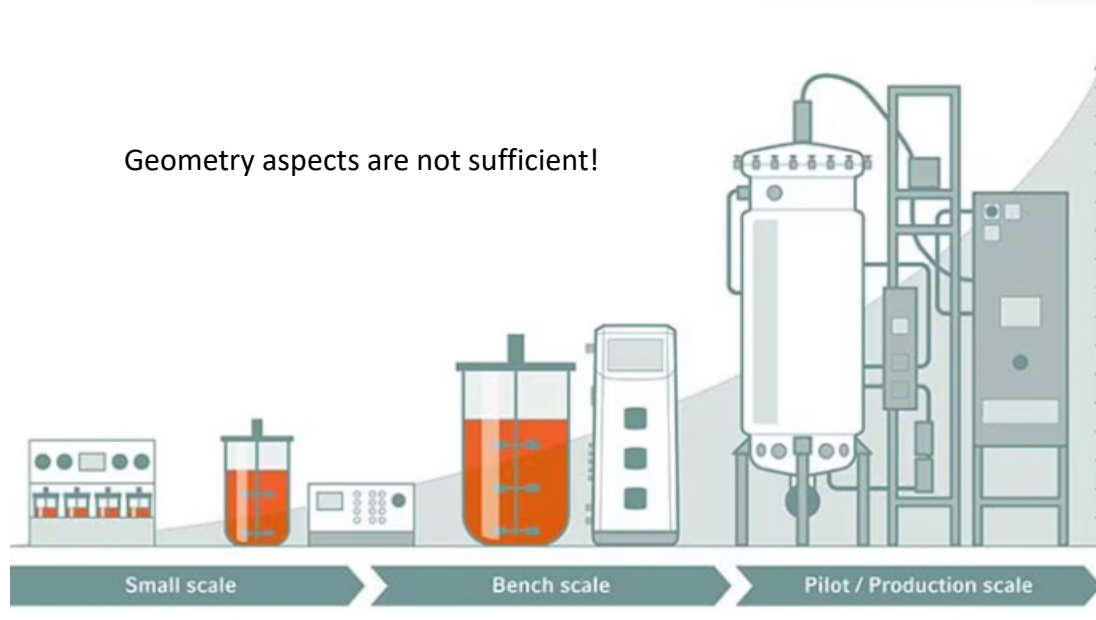
- **Sieving:** a low mesh sieve to break up soft agglomerates
- **Wet Granulation:** Impeller and chopper speed optimization; binder addition rate; wet-mass consistency check
- **Drying:** inlet temperature; air volume; design to minimize attrition
- **Milling + blending:** screen size of millers based on capsule or tablet; check if the initial physical property has improved; avoid excess blending to prevent hygroscopy (timed tests)
- **Capsules:** improved flow and density for consistent fill weight
- **Compression:** optimal hardness, thickness, and disintegration time
- **Packaging:** stability profile under RT and 75% RH; hygroscopy must be considered

03 Reliable scale-up process and solid-liquid reactions

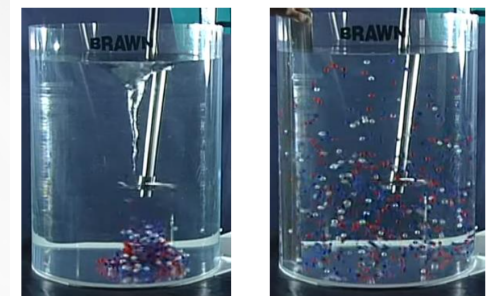
- **Consistent** performance (yield, purity, quality) when moving from lab to industrial scale
- Ideal scale-up = maintaining similar **kinetics** and **spatio-temporal profiles** of concentration and temperature

In solid-liquid reactions, performance depends not only on chemistry, but also on solid suspension, interfacial contact, and mass transfer

Key challenge: mixing rate vs reaction rate



<https://www.eppendorf.com/us-en/lab-academy/applied-industries/bioprocessing/introduction-to-bioprocessing/bioprocessing-scale-up/>



Settled Bed

$N > N_{js}$

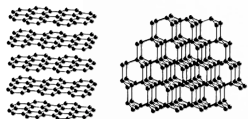
Ensuring $N > N_{js}^{**}$ to maintain consistent interfacial area and mass transfer across all scales

$^{**} N_{js}$ (just suspended speed)

Reliable scale-up $\equiv$ maintaining consistent mixing upon increase in scale

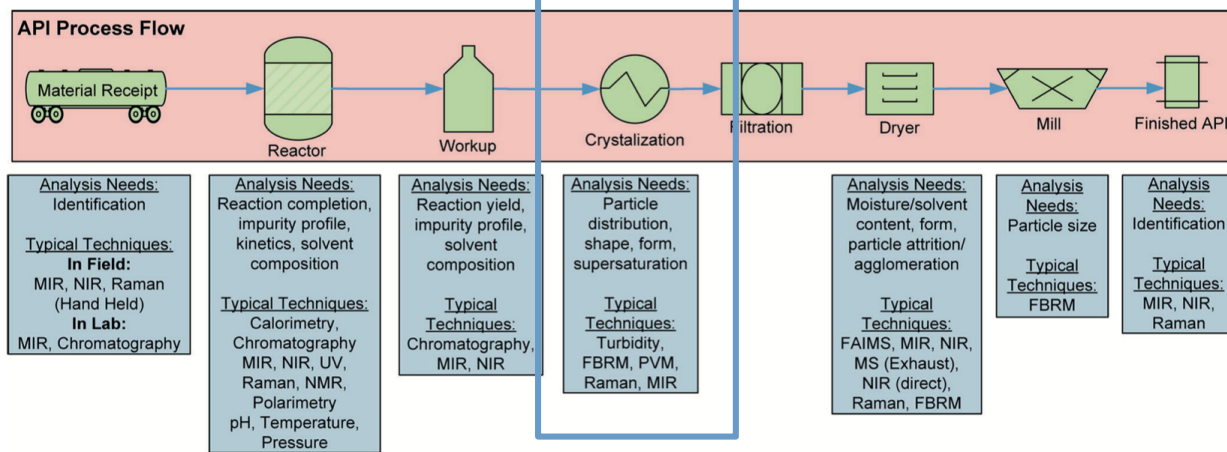
Importance of Drug Substance Properties in Drug Product Process for Solid Dosage Form

Physical Property	Impact
Particle size	Formulation and design considerations, flowability, toxicity, solubility, permeability
Polymorphism (crystal forms)	Solubility, bioavailability, stability
Hygroscopy	Stability, formulation/design considerations, packaging, hydrate formation etc
Bulk density	Flowability, weight variation in capsule filling and compression
Solubility	Bioavailability; crystallization strategy
Melting point	Thermal stability
Compactibility	In tablet formation, density also plays a role



Crystallization is the key unit operation for purity and shape

It determines not only purity, but also the final solid form and manufacturability of the API



Purity



Yield



Size



Polymorphism

Quality requirements for pharmaceutical crystals

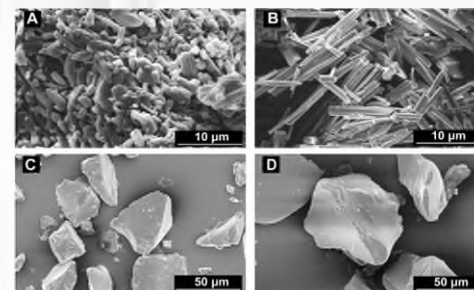


Morphology

It is critical for the drug's clinical performance

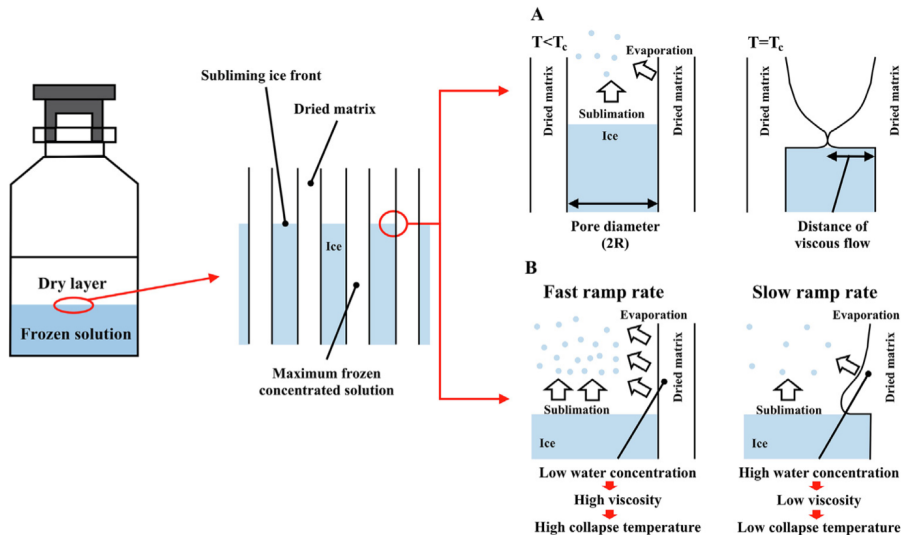
Bioavailability

Risk mitigation

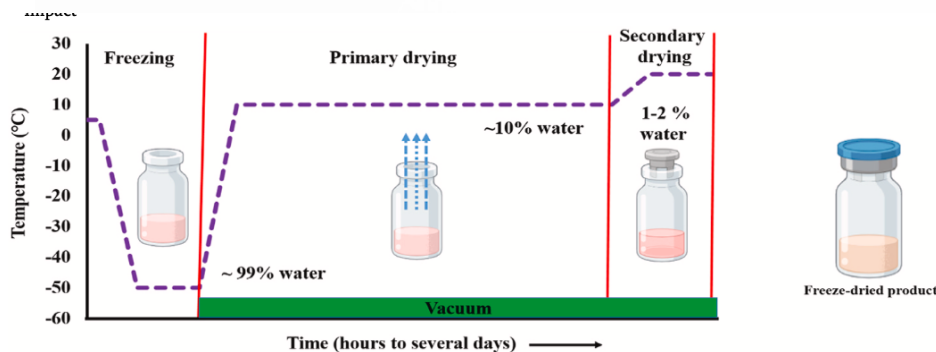


Lyophilization: the most critical unit operation in large molecule drug product process

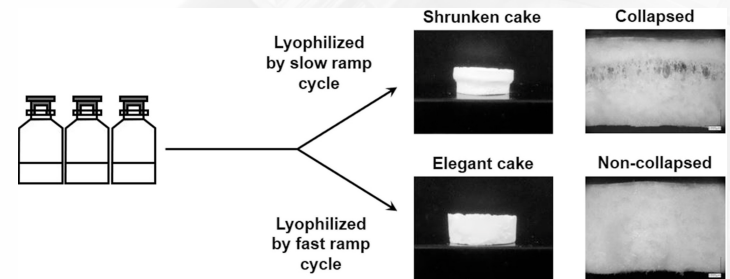
Precise control of heat and mass transfer is required



General overview



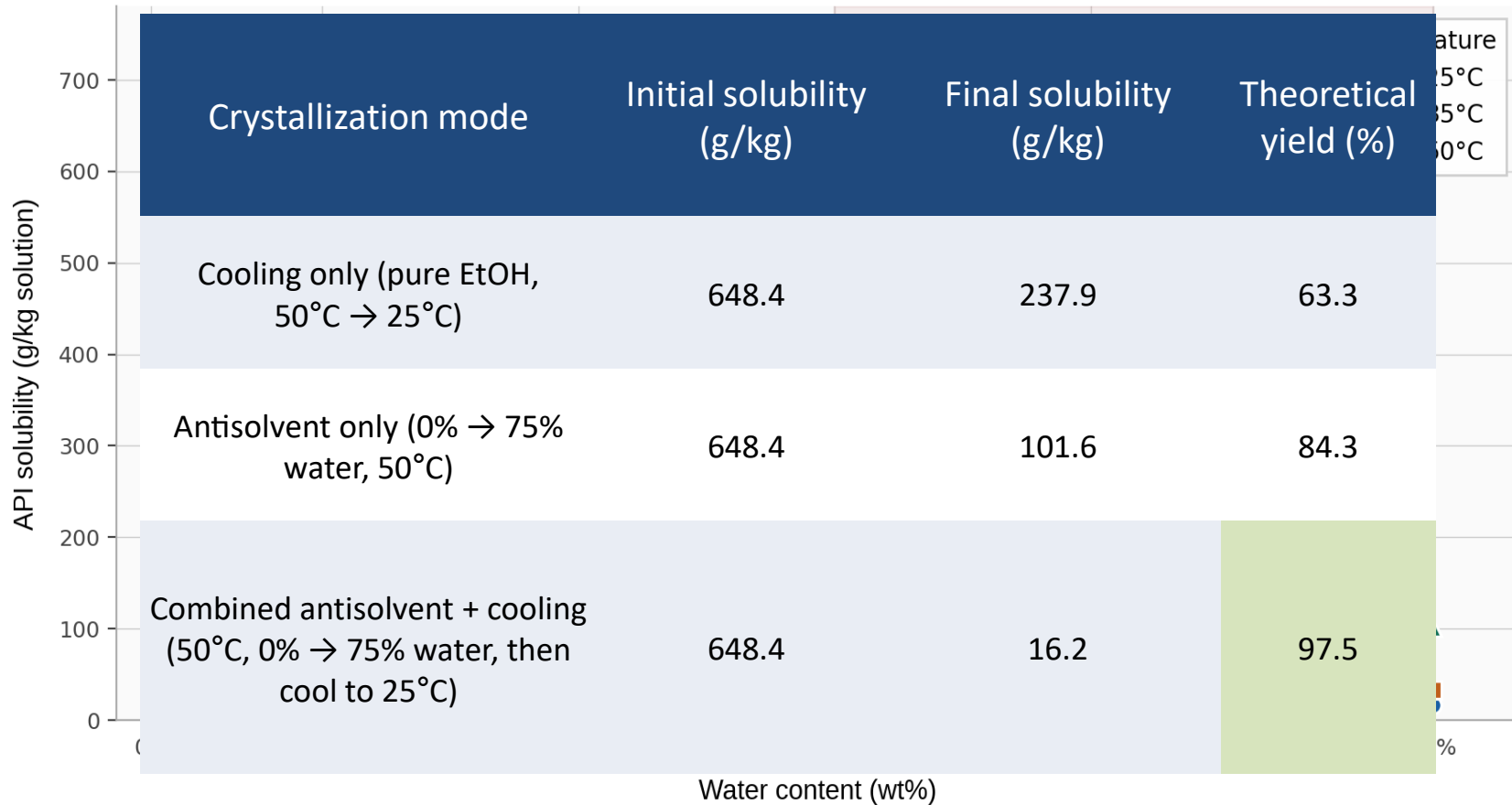
- Complex dynamic solid-liquid-gas system
- Strict temperature constraint: $T < T_g'$
- Risk of cake collapse → product failure
- Scale-up challenges
- Intrinsic instability of biologics



**** More than 50% of approved biopharmaceuticals by FDA and EMA, rely on lyophilization, highlighting its critical importance in drug product manufacturing.**

Crystallization process to maximize the yield given solubility values

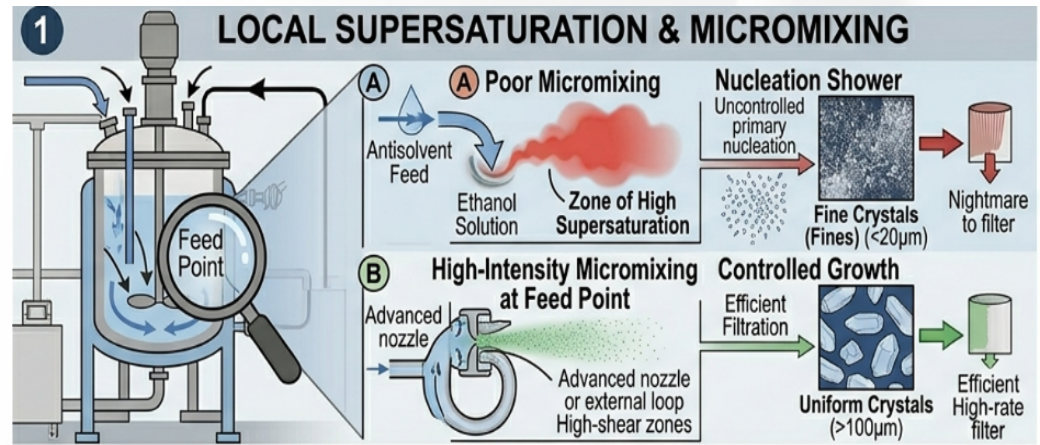
API Solubility in Ethanol–Water Mixtures as a Function of
Temperature and Solvent Composition



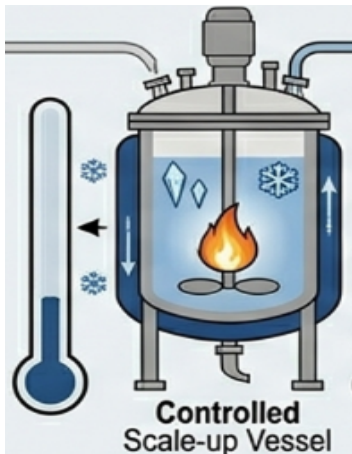
API solubility vs. water content at 25, 35, and 50°C. Shaded region (>40% water) indicates zone of maximum supersaturation generation rate.

Crystallization process to maximize the yield given solubility values

- Controlled water addition rate
- Temperature control
- Hydrates and solvate formation
- Polymorphism check, need to find optimal conditions for the expected polymorph
- Determine the MSZW and LLPS zone, then seed appropriately
- Use PAT to monitor process



Created with Gemini



Created with Gemini

- This is very critical because heat transfer and mixing vary rapidly with scaling; uniform temperature needs to be maintained to prevent unwanted nucleation at regions closer to the vessel wall.
- An optimal impeller size and mixing speed are required to prevent the breaking of crystals from crystal-crystal/impeller collision
- Internal heat exchanger or pumping slurry into an external heat exchanger might be helpful

Key Elements & Unit Operation

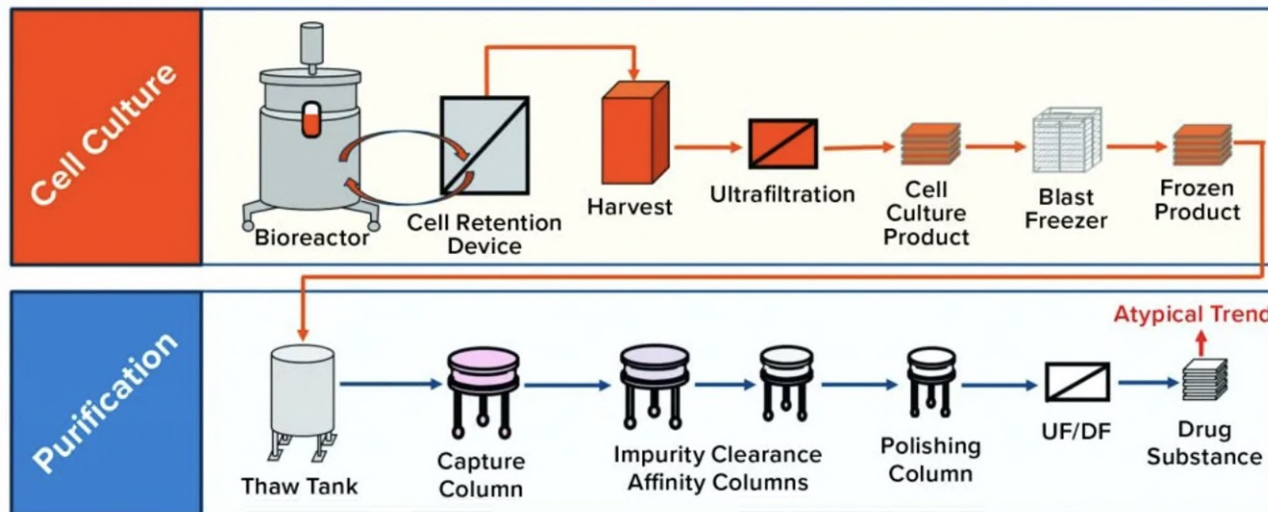
- Cell line and media development
- Core operations

Critical Considerations

- Robustness & Quality
- Control Strategy

Reliable Scale-Up

- Comparable performance, cellular integrity, homogeneity
- Parameter Management
- Mass transfer vs. Shear Stress





THANK YOU

